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Problem 1

(a)

$$z = \frac{R - D_L}{\sigma_L}$$

$$\begin{aligned} E[\text{short}] &= n(R) \\ &= \sigma_L L(z) \\ &= 9.948 \end{aligned}$$

$$\begin{aligned} \text{SS} &= R - D_L + E[\text{short}] \\ &= \boxed{209.948} \end{aligned}$$

(b)

$$\begin{aligned} P(\text{instock}) &= \Phi(z) \\ &= \boxed{0.881} \end{aligned}$$

(c)

$$\begin{aligned} \text{FR} &= 1 - \frac{E[\text{short}]}{Q} \\ &= \boxed{0.99} \end{aligned}$$

(d)

R	$P(\text{InStock})$ (CSL)	FR
600	0.881	0.99
500	0.722	0.971
400	0.5	0.932
300	0.278	0.871
200	0.119	0.79
100	0.039	0.697

- CSL and FR decrease as R decreases
- CSL falls faster
- CSL changes faster with R because it tells us whether a stockout happens at all
- FR tells us how many units are short (compared with Q), grows more slowly

(e)

- $\text{CSL} = P(D_L \leq R)$, the probability of no stockout in a replenishment cycle
  - A cycle that is short by a single unit counts as a complete failure
- $\text{FR} = 1 - E\left[(D_L - R)^+\right]/Q$, the share of demand filled directly from stock
  - A cycle with a stockout can still fill most of its demand; only the units short at the end of the cycle are lost

$$\begin{aligned} \text{FR} &= 1 - E\left[(D_L - R)^+\right]/Q \\ 1 - \text{FR} &= E\left[(D_L - R)^+\right]/Q \\ &= P(D_L > R) \cdot E[D_L - R \mid D_L > R]/Q \end{aligned}$$

Why FR usually exceeds CSL:

$$\underbrace{E\left[(D_L - R)^+\right]}_{\substack{\text{expected shortage} \\ \text{per cycle}}} = \underbrace{P(D_L > R)}_{\substack{\text{stockout probability} \\ =1-\text{ CSL}}} \cdot \underbrace{E[D_L - R \mid D_L > R]}_{\substack{\text{avg. shortage} \\ \text{given a stockout}}}$$

Then:

- $1 - \text{CSL} = P(D_L > R)$
- $1 - \text{FR} = P(D_L > R) \cdot E[D_L - R \mid D_L > R]/Q$

So $\mathrm{FR} > \mathrm{CSL}$ exactly when

$$E[D_L - R \mid D_L > R] < Q$$

- FR exceeds CSL whenever the expected shortfall in a stockout cycle is smaller than the order quantity, i.e. smaller than one cycle's demand
- This is the usual case: a stockout typically falls short by a few units near the end of the cycle (on the order of σ_L), while Q is much larger
- It can reverse when Q is small relative to σ_L (frequent, small orders); then FR can fall below CSL

(f)

Cycle	Demand	Filled	Stockout?
1	10	10	no
2	10	10	no
3	10	10	no
4	100	10	yes

$$\mathrm{CSL} = \frac{3}{4} = 0.75$$

$$\mathrm{FR} = \frac{40}{130} = 0.308$$

Infrequent but large stockouts lead to $\mathrm{CSL} > \mathrm{FR}$

(g)

$$\Phi^{-1}(\mathrm{CSL}) = 1.645$$

$$R = D_L + z(\sigma_L)$$

$$= \boxed{679.141}$$

(h)

$$E[\mathrm{short}] = n(R)$$

$$= 3.546$$

$$\mathrm{SS} = R - D_L + E[\mathrm{short}]$$

$$= \boxed{282.687}$$

(i)

$$L = 2$$

$$s_L = 1.1$$

$$\mu_L = DL = 400$$

$$\sigma_L = \sqrt{L\sigma^2 + D^2 s_L^2} = 277.849$$

$$\mathrm{SS} = \Phi^{-1}(\mathrm{CSL})\sigma_L$$

$$= \boxed{457.021}$$

Lead-time variability raises SS from 279.141 to 457.021 (+63.724%).

(j)

$$L = 2$$

$$D_{L+1} = D(L+1) = 600$$

$$\sigma_{L+1} = \sigma\sqrt{L+1} = 207.846$$

$$z = \Phi^{-1}(\mathrm{CSL}) = 1.645$$

$$S = D_{L+1} + z(\sigma_{L+1}) = \boxed{941.876}$$

(k)

$$\mathrm{SS} = S - D_{L+1} = \boxed{341.876}$$

(l)

In periodic review the safety stock is larger because it covers the uncertainty over $T + L$ time units (T units of time more than in the continuous case)

- Periodic review covers demand over $T + L = 3$ weeks
- Continuous review cover $L = 2$ weeks

Problem 2

(a)

$$z = \Phi^{-1}(0.999) = 3.09$$
$$\sigma_{T+L} = \sigma\sqrt{T+L} = 100.623$$
$$\text{SS} = z\sigma_{T+L} = \boxed{310.949}$$

(b)

$$C_{\text{order}} = S \cdot 52 = \boxed{3016}$$
$$C_{\text{hold}} = H\left(D\frac{T}{2} + \text{SS}\right) = \boxed{31447.463}$$

(c)

	Weekly	Every 2 weeks
SS	310.949	439.748
OUT	1200.949	2219.748
Ordering	3016	1508
Holding	31447.463	55317.507
Total	34463.463	56825.507

- They should not switch
- Ordering $T = 2$ (weeks)
  - 1508 is saved on ordering costs
  - adds 23870.044 to holding costs

Problem 3

(a) What are the optimal order quantity (Q) and reorder point (R) for the backorder and lost sales cases?

$$Q_0 = \text{EOQ} = \sqrt{\frac{2DS}{H}} = 244.949$$

Backorders: $P(D_L > R) = \frac{QH}{bD}$

Lost sales: $P(D_L > R) = \frac{QH}{QH+bD}$

$$Q = \sqrt{\frac{2D(S + bn(R))}{H}}, \quad n(R) = \frac{(200 - R)^2}{400}$$

	Q	R
Backorder	249.136	191.696
Lost sales	248.802	192.037

(b) What do you observe?

- $Q \approx 249$ and $R \approx 192$ in both cases.
- Lost sales gives a slightly lower Q and a slightly higher R

(c) What is your intuitive explanation?

- R is higher with lost sales

  - A backorder is filled from the next delivery and saves some holding cost while a lost sale saves nothing
  - Each shortage costs $h + bD/Q$ instead of bD/Q, so it pays to hold more SS
- Q is lower with lost sales
  - A higher R means a smaller $n(R)$, $Q = \sqrt{2D(S + bn(R))/h}$ decreases
- The difference is small
  - The critical ratios differ only by $hQ \approx 500$ against $bD = 12000$, so $P(\text{stockout})$ decreases